

Silent Vision

Personalized fixed-phrase recognition from a short camera clip with audio disabled.

PROJECT PROFILE

One silent clip. One inspectable phrase decision.

Silent Vision recognizes a small catalog of personalized fixed phrases. It is not open-vocabulary lipreading. Exact accepted text and intent come from the registered catalog.

zh_light_on_hello

你好，请帮我打开灯 -> LIGHT_ON

zh_chat_meal

你吃饭了吗？ -> CHAT_OTHER

en_light_on_hello

Hello, please turn on the light. -> LIGHT_ON

en_chat_meal

Have you eaten? -> CHAT_OTHER

CURRENT STATUS

The user selects the language before recording; the service does not infer it. Selected-language softmax then scores only the enabled phrases for that language. CPU code prepares the mouth clip, and the fixed-phrase Torch model runs on AMD Radeon through ROCm.

RADEON EVIDENCE

Frozen final evaluation: 12/12 phrase top-1, 5/12 known accepted, and 9/10 unrelated clips rejected. Personalized same-speaker scope.

Target users

A deliberate visual input for one user and a small registered phrase catalog.

Creator control input

Another application can map an accepted intent to a capture, cue, or editing action.

Accessible service

A small visual phrase set can supplement other communication channels at a service desk.

Noisy workspace

A deliberate visual command can help when microphone recognition is unreliable.

Privacy-sensitive room

The browser captures video without enabling the microphone or continuous audio recording.

PRODUCT BOUNDARY

Silent Vision answers "which registered phrase best fits this visible mouth sequence?" It does not claim a transcript or cross-speaker generalization. The current source implements recognition, not a downstream creator or device action.

An integrator still defines allowed intents, confirmation rules, retention policy, and failure handling.

Product workflow and safety

One short recording, one catalog decision, and a fail-closed rejection boundary.

01

Capture

User selects zh or en, then records one 2-5 second WebM clip with audio disabled.



02

Preprocess

Decode, detect one face, align, and crop the mouth on CPU.



03

Classify

Run selected-language softmax, phrase logits, and a normalized embedding on Radeon.



04

Gate

Require checkpoint probability and phrase-centroid distance.



05

Route

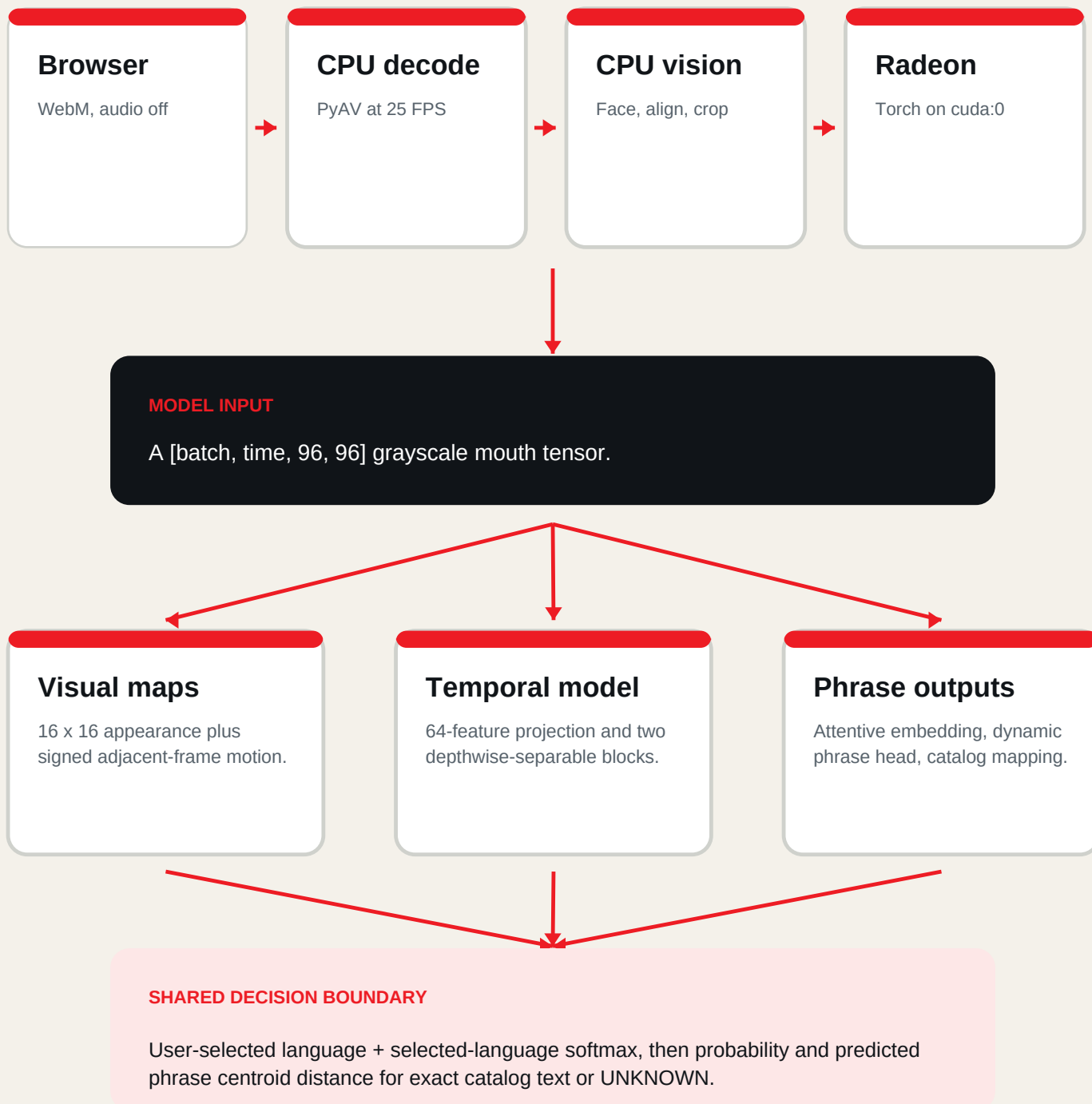
Return exact catalog text or reject the clip as UNKNOWN.

SAFETY RULE

Both calibrated gates must pass. Heuristic UNKNOWN carries no matched phrase text and cannot execute. Top-1 margin is diagnostic only.

System architecture

CPU media preprocessing feeds one fixed-phrase model on AMD Radeon through ROCm.



Model and algorithm

A small temporal model learns phrase classes and a calibrated embedding boundary.

1

Appearance + motion

Downsample to 16 x 16 and compute signed adjacent-frame differences



2

64-feature projection

Learn a compact visual representation for every frame



3

2 temporal blocks

Residual depthwise and pointwise one-dimensional convolutions



4

Embedding + phrase head

Attentive pooling, normalized embedding, dynamic catalog classes

Acceptance

The top phrase must pass the global minimum probability and its maximum cosine distance from the training centroid. Margin is diagnostic only.

Catalog boundary

Selected-language softmax is applied before gating. UNKNOWN is not trained; accepted exact text and intent come from the checkpoint catalog.

AMD Radeon and ROCm

The official path fails closed unless the fixed-phrase Torch model can use ROCm cuda:0.

CPU preprocessing

PyAV decode, 25 FPS resampling, MediaPipe face detection, alignment, and 96 x 96 mouth cropping.

Radeon classification

PyTorch runs visual maps, temporal blocks, attentive embedding, phrase logits, and centroid comparison through ROCm.

REPRODUCTION GUARDS

- `torch.version.hip` must be non-empty.
- `cuda:0` must be visible and accept an allocation.
- A non-empty fixed-phrase checkpoint must be provided.

EVIDENCE STATUS

Official Radeon evidence: 12/12 phrase top-1, 5/12 known accepted, 5/5 accepted known phrases correct, and 9/10 unrelated clips rejected. Inventory, checkpoint, and report lineage are evidentiary and verified.

RUN IDENTITY

PyTorch 2.9.1 | ROCm HIP 7.2 | `cuda:0` | `gfx1100` | 46,405 parameters | checkpoint
`c70d28ae2ed84ee4b2cb0811ebff18870e7b905c1540bc96a782656fba385453`

<https://github.com/fangjixin/silent-vision>